Waseem Raza, PhD

Machine Learning Engineer | AI/ML for Wireless & 5G/6G Systems

Wireless AI | 5G/6G | RAN Intelligence | Network Automation

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LinkedIn | GitHub | Google Scholar | Portfolio

Professional Profile

AI/ML Engineer for Wireless Networks with a PhD in Electrical and Computer Engineering and 7+ years of combined industry and research experience in 5G/6G systems, network intelligence, propagation modeling, mobility, RAN optimization, anomaly detection, and resilient automation. Apply deep learning, transfer learning, domain adaptation, generative modeling, reinforcement learning, and statistical methods to network-assurance, NR-NTN, CSI compression, digital-twin, outage-management, and self-healing problems. Experienced with Python, PyTorch, TensorFlow, scikit-learn, SQL, large-scale network measurements, 3GPP-compliant simulation, and production-oriented ML validation.

Technical Competencies

AI/ML for Wireless: Transfer learning, domain adaptation, adversarial learning, generative modeling, reinforcement learning, anomaly detection, classification, and robust model evaluation

Wireless Systems: LTE/4G, 5G-NR, NR-NTN, mobility, RAN, propagation, cell selection, user timing, network measurements, and service assurance

Network Intelligence: Root-cause analysis, outage detection and compensation, self-healing, optimization, resilience, and intelligent automation

Simulation & Digital Twins: 3GPP-compliant simulation, propagation modeling, synthetic data generation, digital twins, large-scale experimentation, and robustness testing

Programming & Data: Python, SQL, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, visualization, and statistical analysis

Engineering & MLOps: Git, Docker, CI/CD, reproducible experimentation, APIs, cloud analytics, model validation, and production-oriented workflows

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Lead AI/ML development for E911 and network-assurance diagnostics, applying NLP, domain adaptation, and root-cause classification to operational telecom workflows.
- Develop multimodal diagnostic models using work-order text, location information, network context, and operational signals for VoNR and 5G service assurance.
- Design cross-domain validation and distribution-shift evaluation workflows to improve reliability of ML models across heterogeneous network environments.
- Collaborate with network engineers, software teams, and public-safety stakeholders on AI-enabled automation for critical telecommunications operations.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Graduate Research Assistant **09/2019 - 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Selected Projects & Research

Projects & Applied Research

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*
Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.
Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

TurboRAN Experimental Platform *Research Infrastructure | 2020-2024*
Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.
Tools & Methods: 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

Propagation Models & Digital Twins *Research Project | 2024*
Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.
Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

AI for Telecom Network & Security Optimization *Certification Project | 2026*
Applied AI concepts to telecommunications network optimization, security analysis, and customer-experience improvement through a structured specialization project. Framed telecom use cases around operational data, AI-assisted analysis, optimization opportunities, and practical service-improvement recommendations.
Tools & Methods: AI for Telecommunications, Network Optimization, Telecom Security, Customer Experience, Python, TensorFlow, Operational Analytics, Service Improvement

4G Cellular Network Fundamentals Lab *Certification Project | 2026*
Consolidated LTE/4G architecture, radio-access and core-network roles, mobility, and service-flow concepts into a structured cellular-network analysis project. Connected protocol and architecture fundamentals with practical interpretation of end-to-end telecommunications behavior.
Tools & Methods: 4G Networks, LTE, RAN, EPC Concepts, Mobility, Radio Resource Concepts, Cellular Networks, Telecommunications

5G + Edge + IoT + AI Architecture Analysis *Certification Project | 2026*
Evaluated how 5G connectivity, edge computing, IoT, and AI interact in service architectures and enterprise deployment scenarios. Analyzed use-case requirements, edge placement, IoT data flows, AI processing layers, and business/operational tradeoffs for connected intelligent systems.
Tools & Methods: 5G, Edge Computing, IoT, Artificial Intelligence, Architecture Analysis, Use-Case Evaluation, Data Flows, Deployment Tradeoffs

Publications

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *2025*
Sensors 2025

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *2024*
IEEE PIMRC

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *2024*
IEEE SmartNets

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) *2022*
IEEE Network

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *2022*
IEEE WCNC

Unification of RF Energy Harvesting Schemes under Mixed Rayleigh-Rician Fading Channels *2020*
AEÜ - International Journal of Electronics and Communications

Certifications & Professional Development

1. AI Foundations & Industry Overview for Telecommunication — Coursera | Completed | June 19, 2026 | Certificate

Focus: Introduces AI and machine-learning concepts in the telecommunications domain, connecting network data with automation, optimization, predictive analytics, and operational decision support. Reviews how AI can support network planning, service assurance, and customer-facing workflows.

Tools & Skills: telecom AI/ML use cases, network analytics, automation, optimization, predictive modeling concepts, operational intelligence

2. Network and Security Optimization in Telecommunication — Coursera | Completed | June 20, 2026 | Certificate

Focus: Covers telecommunications network optimization together with security considerations that affect service performance and reliability. Emphasizes performance analysis, operational improvement, risk awareness, and protecting network services while maintaining efficient operation.

Tools & Skills: telecom network optimization, network performance, security concepts, service reliability, operational analysis, risk awareness, network improvement

3. Customer Experience in Telecommunication — Coursera | Completed | June 28, 2026 | Certificate

Focus: Covers customer-experience management in telecommunications by linking service quality, customer journeys, operational performance, and analytics to satisfaction and retention. Emphasizes identifying experience gaps through telecom service data and performance indicators.

Tools & Skills: customer experience, quality-of-experience concepts, customer journey analysis, telecom analytics, churn and retention concepts, service improvement

4. 4G Network Fundamentals — Institut Mines-Télécom / Coursera | Completed | August 2, 2026 | Certificate

Focus: Covers LTE/4G architecture from radio access through core-network functions, including cellular-network evolution, mobility, resource management, and end-to-end service operation. Emphasizes how major network components interact to deliver reliable broadband connectivity.

Tools & Skills: LTE/4G architecture, RAN and EPC concepts, mobility and handover, radio resource management, cellular networking, network-service fundamentals

5. Business Considerations for 5G with Edge, IoT, and AI — EDUCBA / Coursera | Completed | February 18, 2026 | Certificate

Focus: Examines how 5G capabilities combine with edge computing, IoT, and AI to enable new enterprise and telecommunications services. Emphasizes use-case evaluation, business value, deployment considerations, and technology tradeoffs across connected intelligent systems.

Tools & Skills: 5G business cases, edge computing, IoT, AI integration, use-case analysis, deployment tradeoffs, telecom strategy

6. Machine Learning — Stanford Online / Coursera | Completed | March 29, 2020

Focus: Covers foundational machine-learning methods for supervised and unsupervised problems, including regression, classification, model evaluation, regularization, clustering, and practical model-selection considerations. Emphasizes translating data into predictive models while controlling generalization error.

Tools & Skills: supervised learning, regression, classification, regularization, clustering, model evaluation, bias-variance tradeoffs, machine-learning workflows